

Practical No : 7

Tutorial Title: Write a program in C++ to implement String class. Write constructors, destructors, accept functions and display functions.

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Program:

```
#include <iostream>
using namespace std;
class student {
    int rollNo;
    float marks;
public:
    student() {
        rollNo = 0;
        marks = 0.0f; }
    student(int k, float m) {
        rollNo = k;
        marks = m; }
    student(const student &s) {
        rollNo = s.rollNo;
        marks = s.marks;
        cout << "\nCopy Constructor called." << endl; }
    ~student() {
        cout << "Destructor called for Roll No: " << rollNo << endl; }
    void disp() {
        cout << "\nStudent Details :" << endl;
        cout << "Roll No: " << rollNo << endl;
        cout << "Marks: " << marks << endl; }
};

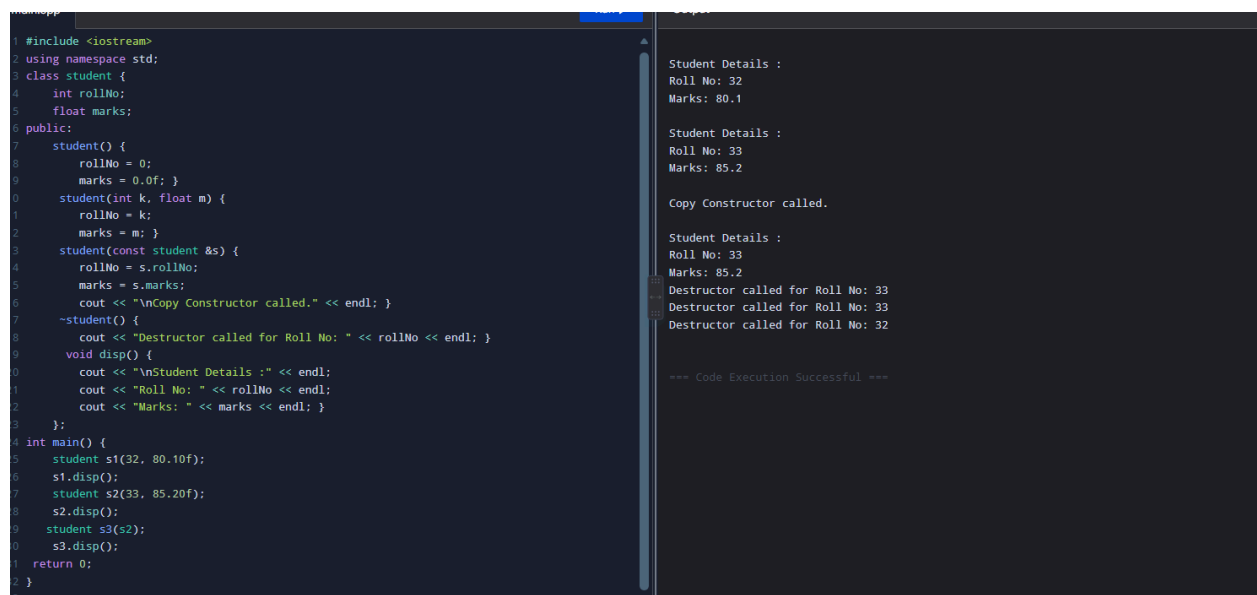
int main() {
    student s1(32, 80.10f);
    s1.disp();
    student s2(33, 85.20f);
    s2.disp();
    student s3(s2);
```

```

        s3.disp();
    return 0;
}

```

Output:



The image shows a C++ IDE with a code editor on the left and an output console on the right. The code defines a `student` class with a default constructor, a parameterized constructor, a copy constructor, and a destructor. The `main` function creates three `student` objects: `s1`, `s2`, and `s3` (created via copy constructor). It calls `disp()` on each and returns 0. The output console shows the execution results, including student details, constructor calls, destructor calls, and a success message.

```

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using namespace std;
class student {
    int rollNo;
    float marks;
public:
    student() {
        rollNo = 0;
        marks = 0.0f; }
    student(int k, float m) {
        rollNo = k;
        marks = m; }
    student(const student &s) {
        rollNo = s.rollNo;
        marks = s.marks;
        cout << "nCopy Constructor called." << endl; }
    ~student() {
        cout << "Destructor called for Roll No: " << rollNo << endl; }
    void disp() {
        cout << "nStudent Details : " << endl;
        cout << "Roll No: " << rollNo << endl;
        cout << "Marks: " << marks << endl; }
};

int main() {
    student s1(32, 80.10f);
    s1.disp();
    student s2(33, 85.20f);
    s2.disp();
    student s3(s2);
    s3.disp();
    return 0;
}

```

```

Student Details :
Roll No: 32
Marks: 80.1

Student Details :
Roll No: 33
Marks: 85.2

Copy Constructor called.

Student Details :
Roll No: 33
Marks: 85.2
Destructor called for Roll No: 33
Destructor called for Roll No: 33
Destructor called for Roll No: 32

=== Code Execution Successful ===

```

